

POWER ELECTRONICS INTRODUCTION/REVIEW

I-1. THE MAIN CONCEPTS OF POWER ELECTRONICS

I-1.1 WHAT IS POWER ELECTRONICS?

Power Electronics (PE) is all about the **efficient *conversion*** of electrical power between various types of source and load.

Power Electronics Is Everywhere



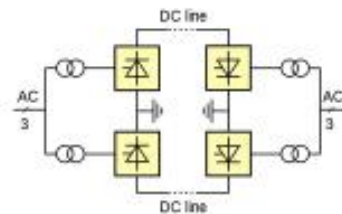
Distributed Power Generation



Power Electronics
Reshape
Electric Power Systems



HVDC Transmission

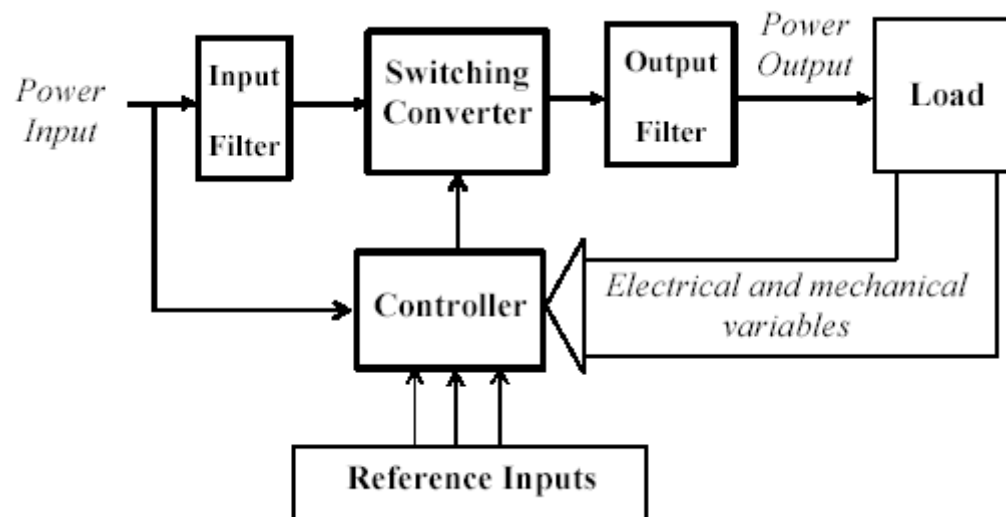


Various Electrical Loads

Conversion of electrical power is carried out **efficiently** by using (primarily) semiconductor switches to vary voltages and current flow. The semiconductors are used in their switching mode (off or on – just like in digital electronics) so that they **do not** dissipate as heat an appreciable amount of the power being converted.

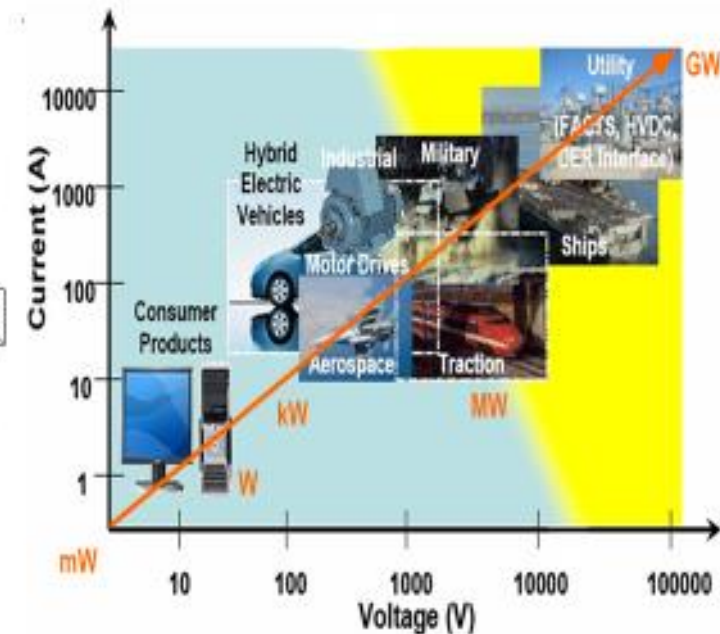
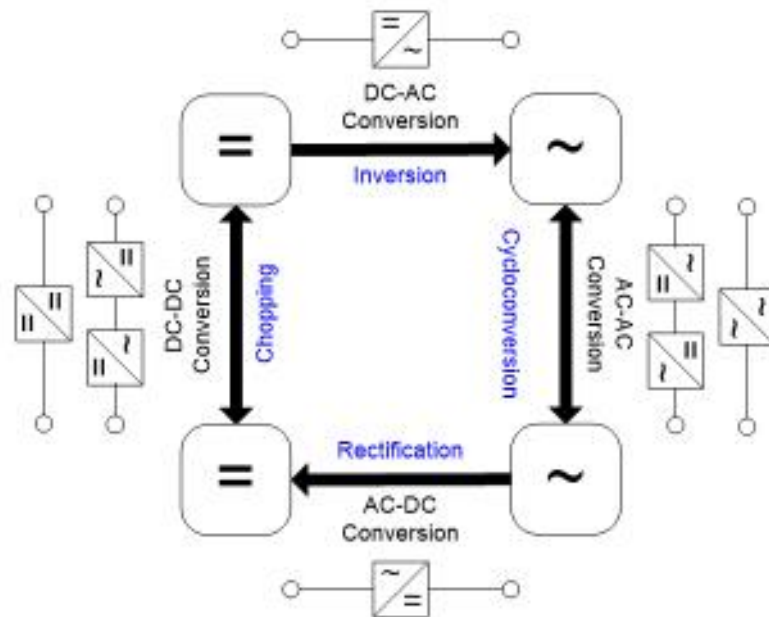
Naturally, it is possible to convert power using semiconductors in their linear mode. Even though this can sometimes be done with an efficiency of up to around 80%, more often than not, it is inefficient and has functional limitations not encountered by switching. This being the case, linear power conversion is not included under the subject area of PE.

A general block-diagram representation of an entire Power Electronics system is shown below.



I-1.2 CLASSIFICATION AND APPLICATIONS

So the idea is to **convert from** some source electrical power which could be *regulated* AC (eg. from the mains), unregulated AC (eg. from a wind-turbine generator), regulated DC (eg. from a computer power supply) or unregulated DC (eg. from a battery), **to** a regulated output electrical power (AC or DC) matched to a desired load.

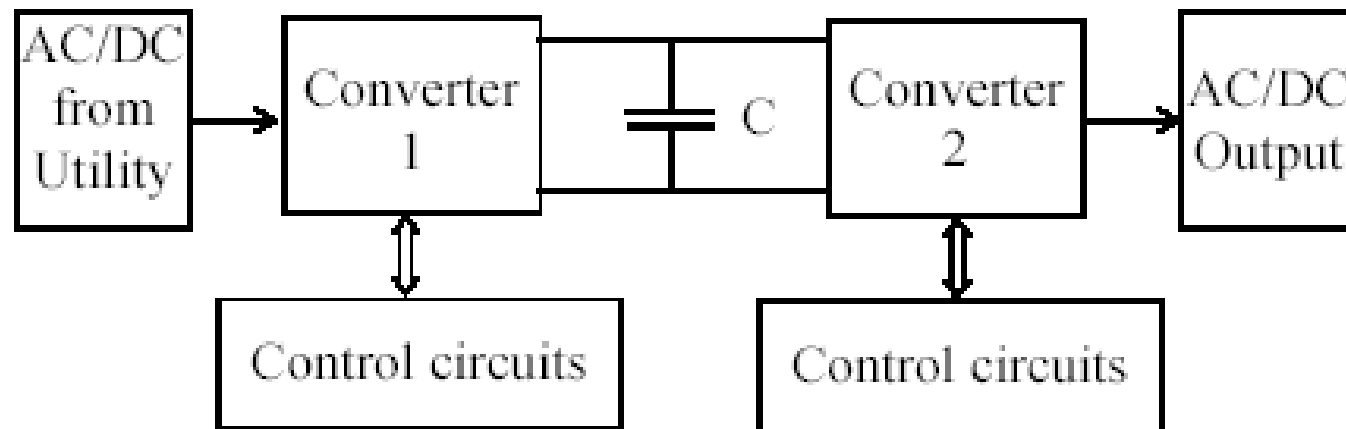


1. PE enables all kinds of efficient power conversion
2. Power rating of Power Converters range from mW to GW

As you might expect, owing to the two fundamental forms of electrical power there are 4 major converter classifications. These being **AC-DC**, **DC-DC**, **DC-AC** and **AC-AC** converters.

While DC-DC and AC-AC converters are simply known as converters, AC-DC and DC-AC converters are also known by alternative names. AC-DC converters are often called *rectifiers*. DC-AC converters are often called *inverters*.

Often, to avoid non-linearity problems or to match load conditions properly, multiple converters of possibly different classes may be required for the overall conversion process (illustrated below). For example many practical AC-AC converters are actually made up of an AC-DC converter followed by a DC-AC converter.



The applications of PE are extremely varied and summarised in the table below.

<i>Residential</i>	<i>Transportation</i>
Refrigeration Space heating Air conditioning Cooking Lighting Power supplies for consumer products Computers	Electric trains, trams, vehicles Battery chargers Subways Conveyor systems Rapid transit systems
<i>Commercial</i>	<i>Utility Systems</i>
Heating, ventilating, airconditioning Refrigeration Lighting Computers & office equipment Uninterruptible power supplies Elevators	High-voltage DC-DC transmission Static VAR compensation Renewable energy sources (wind, PV, fuel cells) Boiler feed systems Energy storage
<i>Industrial</i>	<i>Aerospace</i>
Pumps Compressors Blowers and fans Machine tools Arc and induction furnaces Lighting Welding Process plants	Space vehicle power systems Satellite power systems Aircraft power systems Aircraft controls
	<i>Telecommunications</i>
	Battery chargers Power supplies

I-2. ELECTRONIC SWITCHES

We will now take a quick look at some of the most common semiconductor switching elements used in PE and their basic switching characteristics.

I-2.1 DIODES

You should already know what a diode is and be familiar with the basics of diode characteristics. Figure 2.1 shows the diode and its characteristics.

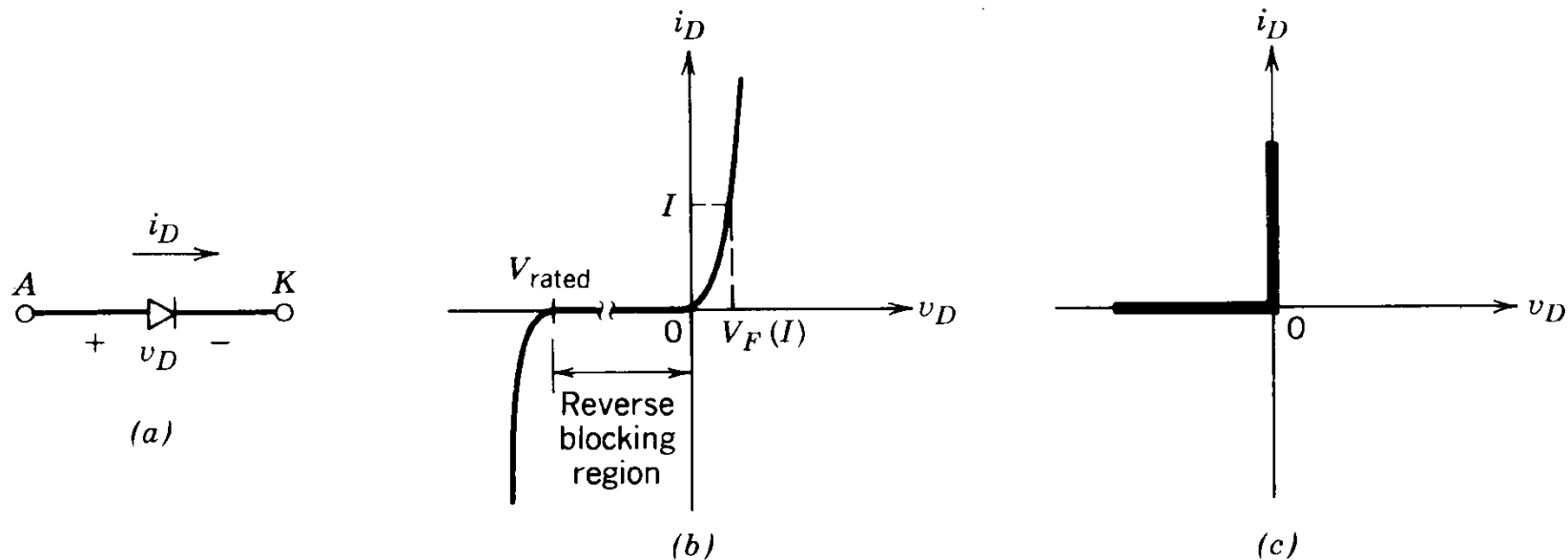


Figure 2.1 Diode symbol and basic characteristics

I-2.2 TRANSISTORS

The relatively specific requirements of PE applications means that the most commonly used transistors are MOSFETs, Insulated Gate Bipolar Transistors (**IGBTs**), and Bipolar Junction Transistors (BJTs). The circuit symbols of these transistors are shown in Figure 2.2

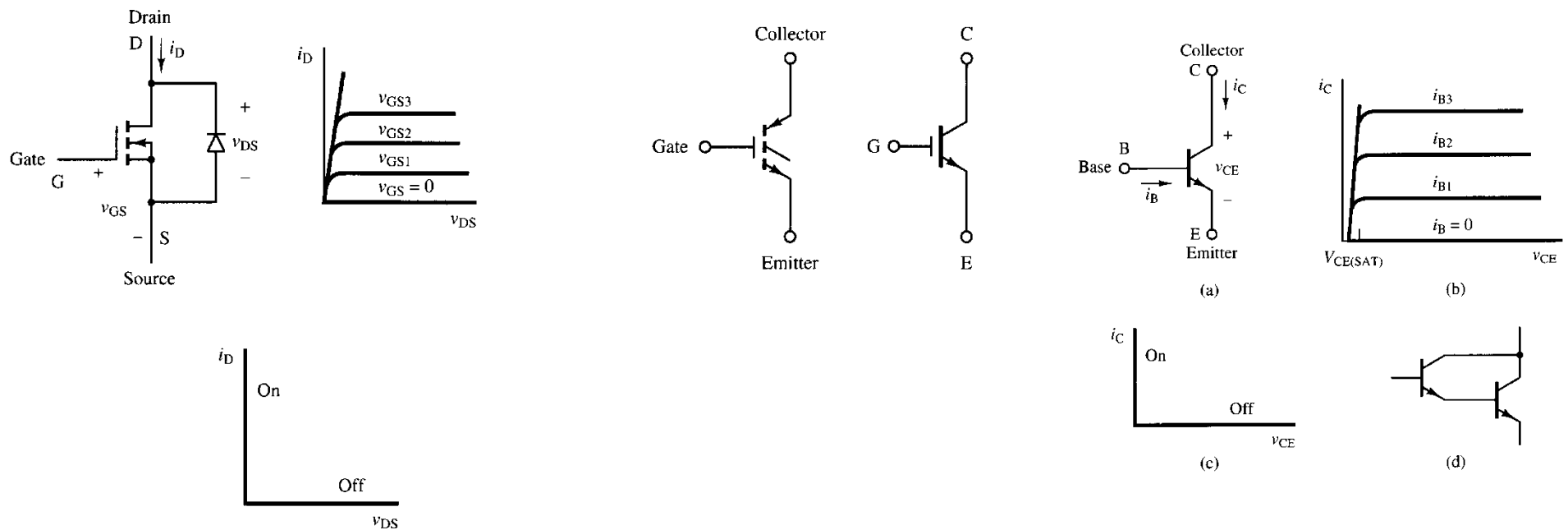


Figure 2.2 Transistors commonly used for PE applications

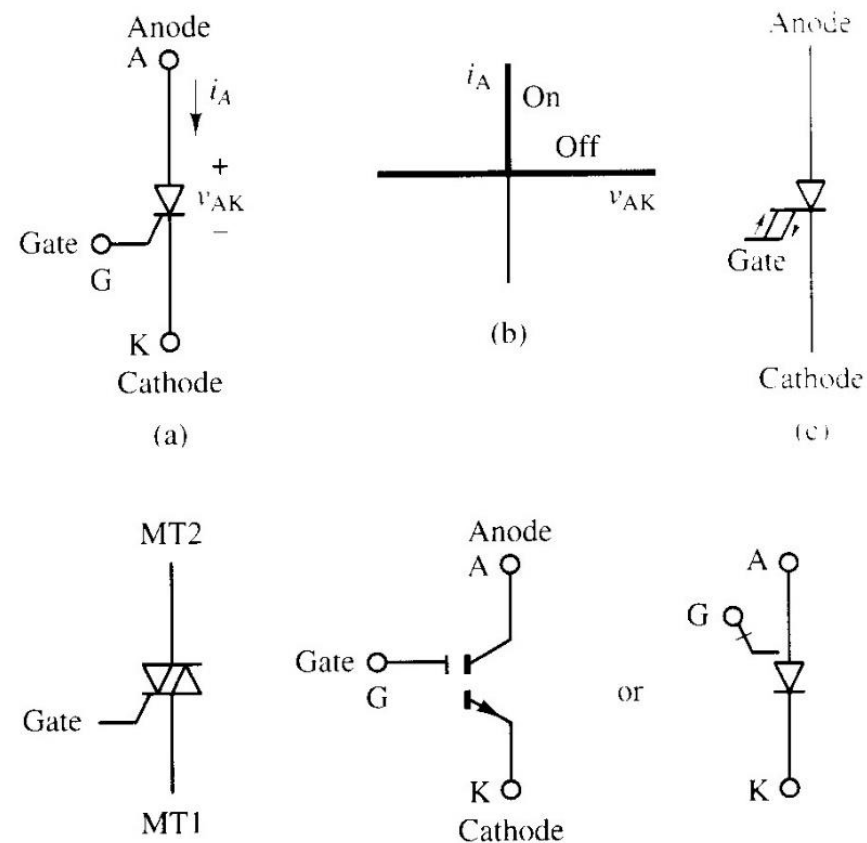
MOSFETs are voltage controlled current devices. The power of the voltage signal into the gate is often small so MOSFETs are usually easier to drive than the likes of BJTs. MOSFETs can not handle as much current as BJTs, but can switch much faster.

The IGBT is a hybrid device that has a MOSFET-style input and a BJT output. That means it is a voltage controlled device that switches faster than BJTs (but still slower than MOSFETs), and can handle larger currents and voltages than MOSFETs. The IGBT has replaced the BJT in most modern PE applications.

BJTs are current controlled current devices. In order to be able to cope with large collector currents, the semiconductor design results in a relatively low current gain. This means the signal into a BJT is sometimes required to have a significant current (in the region of amps). The Darlington BJT arrangement is often used to reduce the required base current. However, Darlington BJTs have slower switching speeds.

I-2.3 THYRISTORS

Thyristors are a group of devices that include the Silicon Controlled Rectifier (**SCR**), the **triac**, the Gate Turn-Off (**GTO**) thyristor, and the MOS-Controlled Thyristor (**MCT**). The circuit symbols for these devices are shown in the Figure below.



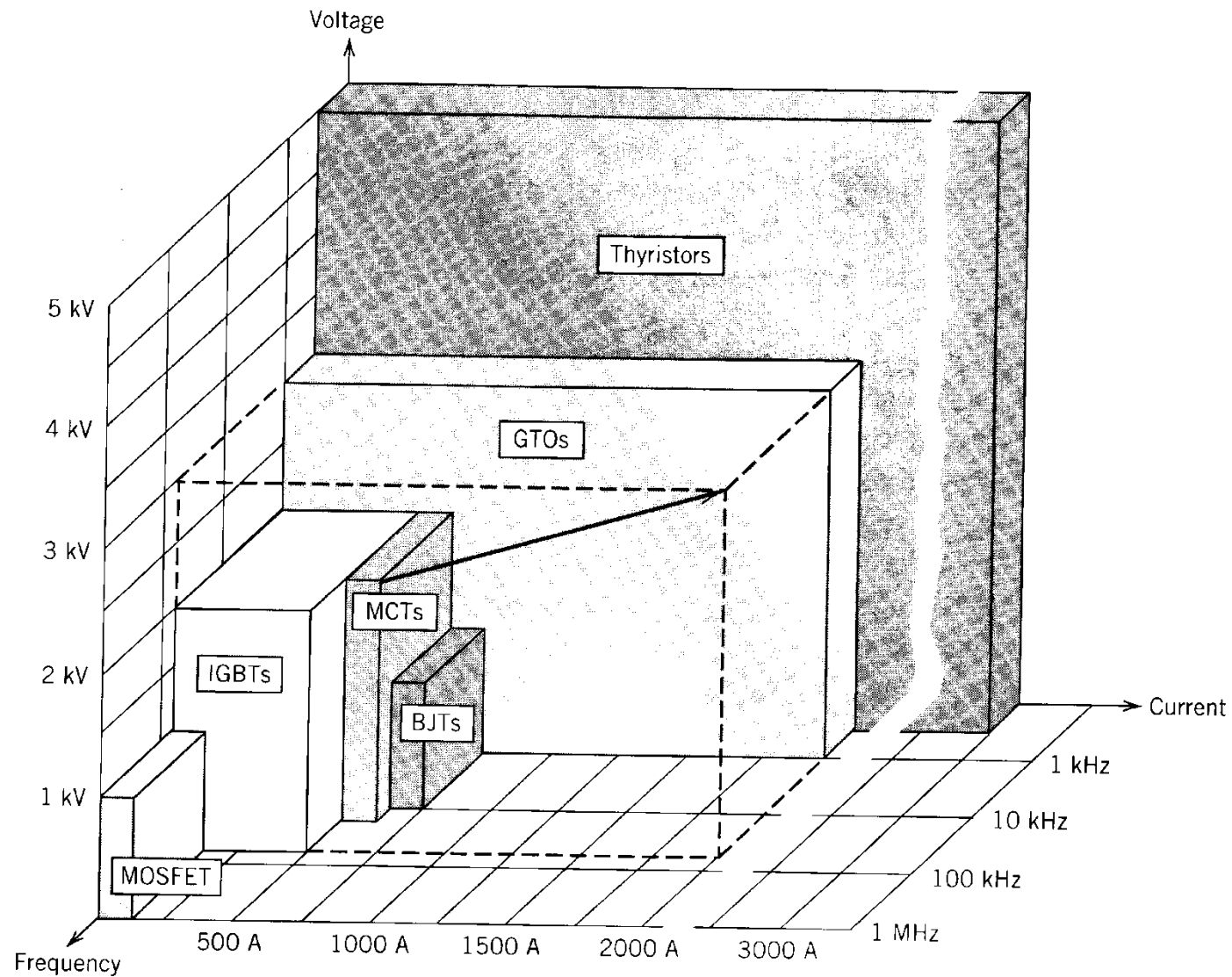
The thyristor group

SCRs and triacs are normally non-conducting when biased in either polarity but can be made to conduct by applying an appropriate signal to the gate terminal (current flow in only one direction for the SCR, whereas the triac can conduct current in both directions). When the current through an SCR or triac falls below a small *holding current*, it will automatically stop conducting. This is the only way to switch these two devices off.

GTOs and MCTs can be switched on but can be switched off as well, making them more versatile in some applications. GTOs require a rather high gate pulse current to turn them off, but can handle extremely large currents and voltages. MCTs require much less gate signal power to turn them off, but are not capable of handling as high currents and voltages.

Thyristors as a family of devices used in PE applications tend to be quite slow in their switching characteristics, but can handle relatively high currents and voltages.

This Figure summarises the main switching parameters for the devices looked at so far.



PE switching device capabilities.

I-2.4 SWITCH SELECTION

Choosing the right switch for any particular application depends on many factors such as load current and voltage, electrical source form, desired switching speed, economics, and circuit complexity, to name a few.

Many of these issues tend to work themselves out when the application is firmly established and previously designed circuits can be considered.

I-2.5 CAD & CIRCUIT SIMULATION

In order to avoid having to physically build PE circuits we have initially designed and to help test out design ideas, circuit simulation software is often used. One of the most commonly used circuit simulation ‘engines’ is called SPICE (Simulation Program with Integrated Circuit Emphasis) and is the foundation of commercial electronics simulation/design software such as LT-SPICE, OrCAD and Altium.